STANDALONE THEORETICAL NOTE · SECTION 14.1

Point-Like Structures in 4D Space

Resolving the Electron Size Paradox via 4D Funnel Foreshortening

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ABSTRACT & READER POSITIONING

This note is written to be read completely on its own, isolating a single geometric mechanism: how a Compton-scale wave loop circulating at the speed of light inside a 4th-dimensional funnel appears point-like when viewed in three-dimensional space. In flat 3D space, a one-way circulation of light carrying rest energy $m_e c^2$ and spin $\hbar/2$ must have radius $R_{\text{flat}} = \hbar/(2m_e c) \approx 1.93 \times 10^{-13}$ m, conflicting with LEP e^+e^- collision bounds ($R_{\text{LEP}} < 1.2 \times 10^{-19}$ m) by six orders of magnitude. When the circulating wave instead pushes into a 4th dimension to form a funnel that slows local clocks by $\alpha \ll 1$ and stretches local space by $1/\alpha$, the apparent 3D throat radius shrinks quadratically as $R_{\text{outside}} = \alpha^2\hbar/(2m_e c)$. For $\alpha \leq 8 \times 10^{-4}$, the 3D footprint drops below the LEP limit while the internal momentum blueshift ($p_{\text{inside}} \propto 1/\alpha$) keeps spin $\hbar/2$ exact.

Established in This Note	Open Verification Gates After This Note
Exact three-scale splitting in a conformal 4D funnel (R_{flat}, $R_{\text{inside}} = \alpha R_{\text{flat}}$, and $R_{\text{outside}} = \alpha^2 R_{\text{flat}}$).	Full momentum-transfer scattering calculation $F(q^2)$ connecting collider cross-sections to R_{outside}.
Threshold clock depth $\alpha \leq 8 \times 10^{-4}$ bringing a Compton loop below the 1.2×10^{-19} m LEP contact bound.	Curved-space electromagnetic dressing verifying that the Bohr magneton $\mu_B = e\hbar/(2m_e)$ does not shrink with the throat.
Exact cancellation between proper lever-arm compression (α) and momentum blueshift ($1/\alpha$), preserving $J = \hbar/2$.	Dynamical stability of the one-way spinning mode under simultaneous energy and spin conservation.

1. The 100-Year Size Paradox of Ring Electrons

A recurring physical picture of the electron—spanning classical ring-electron proposals, Dirac's *zitterbewegung* analysis, and geometric algebra formulations **Hestenes (1990)**—models the particle as a massless wave packet circulating at the speed of light c around a closed loop. For light running one way around a ring, energy, spin angular momentum, and radius are locked by Spin = (Energy × Radius)/c. Fixing the electron's externally measured mass-energy $E = m_e c^2$ and intrinsic spin $J = \hbar/2$ in flat three-dimensional space immediately pins the ring's radius to half the reduced Compton wavelength:

$$R_{\text{flat}} = \hbar / (2m_e c) \approx 1.93 \times 10^{-13}\ \text{m}, \qquad R_{\text{LEP}} < 1.2 \times 10^{-19}\ \text{m}.$$

While R_{flat} naturally accounts for the electron's quantum phase cadence, contact-interaction searches in high-energy e^+e^- collisions at LEP **L3 Collaboration (2000)** constrain the electron's hard interaction radius to $R_{\text{LEP}} < 1.2 \times 10^{-19}$ m. That conflict rests on an unstated assumption: that space inside the loop is flat, forcing the oscillation scale, internal path length, and external 3D radius to be identical.

2. Why a 4D Funnel Bends Light and Stretches Space

In a 4D geometric picture, a photon's propagation involves a local push into the 4th dimension. An open photon traveling in a straight line never returns to the same spot, so its dent relaxes behind it. A closed photon circulating around a loop continuously re-pushes the same region before the background can relax, carving a persistent funnel. Inside that funnel, clocks tick slower toward the throat at fraction $\alpha \in (0, 1]$ relative to a far-away clock ($\alpha_\infty = 1$), bending the wave into a circular orbit like a banked optical lens. When the 4D push splits equally between slowing time and stretching space, proper distances down in the throat stretch by $\psi^2 = 1/\alpha$:

$$ds^2 = -\alpha(R,z)^2 c^2 dt^2 + \alpha(R,z)^{-2}(dR^2 + dz^2 + R^2 d\varphi^2), \qquad L_{\text{inside}} = 2\pi R_{\text{inside}} = 2\pi R_{\text{outside}} / \alpha.$$

For light moving at local speed c ($ds^2 = 0$), traversing a path stretched by $1/\alpha$ while local clocks tick at rate α slows the apparent angular rate seen from outside by two powers of α, giving $d\varphi/dt = \alpha^2 c / R_{\text{outside}}$ (optical index $n = \psi^2/\alpha = 1/\alpha^2$).



Figure 1: A Compton-frequency light ring orbiting deep inside a 4D funnel ($\alpha \ll 1$) travels a stretched proper circumference $2\pi R_{\text{inside}}$, but projects onto the external 3D slice as a point-like throat of radius $R_{\text{outside}} = \alpha^2 R_{\text{flat}}$.

3. The Foreshortening Law: $R_{\text{outside}} = \alpha^2 R_{\text{flat}}$

Hold fixed the two quantities that the outside world measures for an isolated electron: its asymptotic mass-energy $E = m_e c^2$ and its canonical spin $J = \hbar/2$. Together, these lock the apparent circulation cadence measured on a far-away clock to $\omega = E/J = 2m_e c^2/\hbar$. Equating $d\varphi/dt = \alpha^2 c / R_{\text{outside}}$ to ω yields the foreshortening law:

$$\boldsymbol{R_{\text{outside}} = \alpha^2 \cdot \hbar / (2m_e c) = \alpha^2 R_{\text{flat}}, \qquad R_{\text{inside}} = \alpha \cdot \hbar / (2m_e c) = \alpha R_{\text{flat}}.}$$

- **The External Quantum Cadence ($R_{\text{flat}} = \hbar/(2m_e c) \approx 1.93 \times 10^{-13}$ m):** Set by the oscillation rate ω on far-away clocks where $\alpha \to 1$, governing low-energy de Broglie wave interference and atomic orbitals.
- **The Internal Proper Radius ($R_{\text{inside}} = \alpha R_{\text{flat}}$):** The stretched circumference the trapped photon actually travels down inside the funnel throat.
- **The Apparent 3D Throat Radius ($R_{\text{outside}} = \alpha^2 R_{\text{flat}}$):** The foreshortened cross-section seen from three-dimensional space, which shrinks with the *square* of the funnel's clock factor α.

4. Quantitative Scaling Across Funnel Depths

Funnel Regime	Clock at Ring (α)	Apparent 3D Size (R_{outside})	Internal Path (R_{inside})	Comparison to Experiment
Flat 3D loop	1.00	1.9×10^{-13} m	1.9×10^{-13} m	$1.6 \times 10^{6}\times$ above LEP limit
Gentle clock well	0.63	7.7×10^{-14} m	1.2×10^{-13} m	Still Compton-sized in 3D
Short-reach funnel	0.086	1.4×10^{-15} m	1.7×10^{-14} m	Shrunk by ~135× in 3D
Long-reach funnel	0.0044	3.7×10^{-18} m	8.5×10^{-16} m	Within ~30× of LEP limit
Point-like threshold	**$\leq 8 \times 10^{-4}$**	**$\leq 1.2 \times 10^{-19}$ m**	**$\leq 1.5 \times 10^{-16}$ m**	**Passes LEP collider bound**
Deep bending funnel	10^{-5}	1.9×10^{-23} m	1.9×10^{-18} m	$10^{4}\times$ below LEP bound

5. Why Spin (ħ/2) Does Not Shrink with the Throat

If the ring's apparent radius shrinks by $\alpha^2 \lesssim 10^{-6}$, why doesn't the particle's spin angular momentum $J = R \times p$ also shrink by 10^{-6}? Because a slow clock at the bottom of the funnel implies that the trapped light is correspondingly blueshifted inside the throat. When the outside world measures a redshifted mass-energy $E = m_e c^2$, a local observer at clock rate α down at the ring measures a local proper energy $E_{\text{inside}} = m_e c^2/\alpha$ and local proper momentum $p_{\text{inside}} = E_{\text{inside}}/c = m_e c/\alpha$. Multiplying the proper lever arm by the proper momentum cancels α identically at every depth, closely analogous to a "bag of gold" throat **Wheeler (1955)**:

$$J = R_{\text{inside}} \cdot p_{\text{inside}} = (\alpha \cdot \hbar / 2m_e c) \cdot (m_e c / \alpha) = \hbar / 2.$$

6. Caveats, Verification Gates & Conclusion

Separating the 3D throat size from the Compton cadence resolves the geometric objection and isolates three concrete verification targets: (1) computing full momentum-transfer scattering cross-sections $F(q^2)$ on the funnel metric to verify how high-energy probes sample R_{outside}; (2) verifying that curved-space electromagnetic dressing preserves the external Bohr magneton $\mu_B = e\hbar/(2m_e)$ across the funnel's outer reach rather than shrinking with the throat; and (3) enforcing simultaneous spin and energy conservation on the one-way spinning mode. Viewed from three dimensions, a Compton wave loop deep inside a 4D funnel naturally foreshortens into a point.

References

1. L3 Collaboration (M. Acciarri *et al.*), "Search for New Phenomena in Fermion-Pair Production at LEP," *Phys. Lett. B* **489**, 81–92 (2000); contact-interaction bound $R_e < 1.2 \times 10^{-19}$ m (1.2×10^{-17} cm).
2. J. A. Wheeler, "Geons," *Phys. Rev.* **97**, 511–536 (1955); *Geometrodynamics* (Academic Press, New York, 1962).
3. D. Hestenes, "The Zitterbewegung Interpretation of Quantum Mechanics," *Found. Phys.* **20**, 1213–1232 (1990).